

Augmented Intelligence in Technology-Assisted Review Systems (ALTARS 2026)

Giorgio Maria Di Nunzio
Department of Information
Engineering, University of Padua
Padova, Italy
giorgiomaria.dinunzio@unipd.it

Evangelos Kanoulas
University of Amsterdam
Amsterdam, The Netherlands
E.Kanoulas@uva.nl

Prasenjit Majumder
Dhirubhai Ambani Institute of
Information and Communication
Technology
Gandhinagar, India
prasenjit_t@isical.ac.in

Abstract

The ALTARS 2026 workshop explores recent advances and open challenges in Technology-Assisted Review (TAR) systems and their application to large-scale, high-recall retrieval across the Web. TAR systems, originally designed for domains such as legal discovery and systematic literature review, are now increasingly relevant to Web environments characterized by heterogeneous, dynamic, and multilingual information. As the Web continues to expand through user-generated and AI-produced content, ensuring transparency, reliability, and fairness in automated review processes has become a central research challenge.

This workshop aims to bring together researchers and practitioners from information retrieval, Web science, artificial intelligence, and data governance to discuss how TAR methodologies can support trustworthy and scalable information access. Topics include intelligent retrieval, human-in-the-loop learning, explainable and responsible AI, and the integration of large language models and knowledge graphs in review workflows. Through paper presentations, keynotes, and interactive sessions, ALTARS 2026 seeks to foster interdisciplinary collaboration and to identify future directions for developing transparent, fair, and adaptive Web-scale review systems.

CCS Concepts

• **Information systems** → **Information systems applications; Retrieval models and ranking; Retrieval tasks and goals; Evaluation of retrieval results.**

Keywords

Systematic Reviews, High Recall Systems, Document Prioritization

ACM Reference Format:

Giorgio Maria Di Nunzio, Evangelos Kanoulas, and Prasenjit Majumder. 2026. Augmented Intelligence in Technology-Assisted Review Systems (ALTARS 2026). In *Companion Proceedings of the ACM Web Conference 2026 (WWW Companion '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3774905.3795597>



This work is licensed under a Creative Commons Attribution 4.0 International License. [WWW Companion '26, Dubai, United Arab Emirates](https://creativecommons.org/licenses/by/4.0/)
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2308-7/2026/04
<https://doi.org/10.1145/3774905.3795597>

1 Workshop Topics and Themes

The Web has evolved into a complex information ecosystem where ensuring the transparency, reliability, and accessibility of content is a major challenge. Technology-Assisted Review (TAR) systems, originally developed for high-recall retrieval tasks such as legal discovery and systematic literature reviews, are increasingly relevant for managing large and dynamic Web-scale data. In an age of information overload and widespread generative content, TAR methods provide a systematic approach to identifying, ranking, and validating relevant information through iterative human-AI collaboration. Integrating TAR principles with emerging Web technologies such as retrieval-augmented generation, knowledge graphs, and multilingual search opens new avenues for building trustworthy and adaptive information systems that can operate at Web scale.

The timing of this workshop is particularly appropriate given the rapid integration of large language models, semantic retrieval, and human-in-the-loop workflows across the Web. By examining TAR through the lens of Web intelligence and responsible AI, the workshop aims to address critical challenges such as explainability, bias mitigation, and scalability in open information environments. ALTARS 2026¹ will serve as a forum for the Web research community to explore how TAR frameworks can enhance Web content curation, facilitate evidence-based information access, and support the development of fair and accountable AI systems. The workshop will thus provide a bridge between information retrieval, Web science, and ethical AI, promoting interdisciplinary dialogue and advancing the collective understanding of high-recall retrieval in the context of the evolving Web.

2 Previous Editions and Evolution of the Workshop

The first edition of the ALTARS workshop was successfully held at ECIR 2022² [4]. During (and after) that workshop, there was a lively discussion about open questions that still need to be addressed and clarified in this research area. The second edition witnessed an even larger participation at ECIR 2023³ [5] with around 20 participants on-site and 10 online. The aim of the second edition was to study innovative approaches to fathom the effectiveness of these TAR systems. The third edition witnessed the same participation at ECIR 2024⁴ [11] with around 20 participants on-site and 10 online. The

¹<https://altars2026.dei.unipd.it/>

²<https://altars2022.dei.unipd.it/>

³<https://altars2023.dei.unipd.it/>

⁴<https://altars2024.dei.unipd.it/>

aim of the second edition was to study innovative approaches to fathom the effectiveness of these TAR systems.

The ALTARS workshop series has evolved significantly since its origin, reflecting the growing intersection between high-recall retrieval, TAR systems, and recommender systems across various domains. Early editions established a strong foundation by exploring theoretical advancements, evaluation methodologies, and real-world applications in areas such as legal AI, medical research, and systematic reviews. Over time, the workshop has expanded to incorporate emerging technologies, including retrieval-augmented generation (RAG), large language models (LLMs), and explainable AI, while also exploring how recommender system techniques can enhance TAR workflows through adaptive ranking, user feedback loops, and content-based filtering.

Building on these successful editions, ALTARS 2026 will further broaden its focus by exploring Technology-Assisted Review (TAR) systems in the context of the Web. This shift reflects the growing importance of TAR methodologies for large-scale, heterogeneous, and dynamic information environments. Insights from earlier editions have highlighted the need for stronger connections between retrieval models, explainable AI, and responsible information governance—topics that align naturally with The Web Conference’s themes. The 2026 edition will therefore introduce new interdisciplinary discussions on the integration of TAR with Web-scale retrieval, knowledge graphs, and large language models, expanding the workshop’s impact beyond its original focus on high-recall retrieval in specialized domains.

3 Workshop Objectives

Technology-Assisted Review (TAR) systems are a specialized class of intelligent retrieval and recommendation technologies designed to help users efficiently navigate large-scale document collections by prioritizing the most relevant information. Traditionally applied in high-stakes domains such as legal discovery, compliance monitoring, and systematic literature reviews, TAR systems share many foundational challenges with Web-based retrieval and recommendation models, including ranking optimization, interactive relevance feedback, active learning, and bias mitigation.

As the Web community increasingly explores responsible, transparent, and domain-specific applications of AI, TAR systems provide a compelling use case where Web-scale recommendation and retrieval methodologies can enhance decision-making in expert-driven contexts, such as legal analysis, biomedical research, and regulatory compliance. Exploring TAR within the WebConf ecosystem is particularly timely, as it bridges areas including intelligent information retrieval, knowledge-augmented generation, large language models (LLMs), and fairness-aware AI — all central to current Web research.

By bringing together experts from information retrieval, Web intelligence, natural language processing, and recommender systems, ALTARS 2026 aims to foster dialogue on how recommender-inspired techniques can improve TAR pipelines and, conversely, how TAR’s rigorous evaluation frameworks can contribute to more accountable and interpretable recommendation and retrieval systems. The ALTARS workshop will promote interdisciplinary collaboration and knowledge exchange, enabling participants to share

findings, define open challenges, and propose future directions for Web-scale high-recall retrieval. By focusing on precision, transparency, and scalability, the workshop will highlight the intersection of machine learning, human-in-the-loop interaction, and responsible AI, offering a comprehensive view of current challenges and innovative solutions.

For ALTARS 2026, we invite contributions addressing the following innovative themes that connect TAR and Web-scale recommendation research:

- (1) **Web-Scale Applications of Technology-Assisted Review.** Research on applying TAR methodologies to large and dynamic Web environments, including open data discovery, scientific knowledge curation, and digital governance. Contributions may explore AI-assisted systems for identifying, filtering, and validating Web content in domains such as legal, biomedical, and environmental information, as well as combating misinformation and ensuring content integrity.
- (2) **Intelligent and Hybrid Retrieval on the Web.** Studies that combine TAR principles with Web-scale search, knowledge graph reasoning, and content recommendation to improve recall and contextual awareness. Topics may include integrating LLM-based retrieval-augmented generation (RAG) pipelines, semantic search, and adaptive ranking for heterogeneous Web data.
- (3) **Interactive, Explainable, and Human-in-the-Loop TAR.** Research focusing on transparent and interpretable TAR interfaces for Web information retrieval. This includes explainable AI (XAI) techniques for document triage, user-in-the-loop systems that support collaborative curation of Web resources, and uncertainty estimation in AI-driven content prioritization.
- (4) **Responsible and Fair Web Review Systems.** Investigations into fairness, accountability, and transparency in large-scale Web review processes. Relevant topics include bias detection and mitigation in Web content filtering, responsible AI in regulatory or journalistic contexts, and ethical design of TAR systems supporting trustworthy and inclusive Web ecosystems.

4 Relevance to The Web Conference 2026

The ALTARS 2026 workshop is a strong fit for The Web Conference, as it directly addresses the challenges of intelligent information access, content quality, and trustworthiness on the modern Web. The workshop connects key areas of interest to the Web community, including large-scale information retrieval, human-AI collaboration, knowledge management, and ethical AI. By focusing on Technology-Assisted Review (TAR) systems as a framework for high-recall retrieval and adaptive decision support, the workshop contributes to ongoing discussions on how to make Web-based systems more transparent, interpretable, and reliable.

ALTARS 2026 also complements several established themes within the conference, such as Web search and mining, Web science, and responsible Web technologies. Its emphasis on explainability, fairness, and accountability aligns closely with current efforts to ensure that AI systems deployed on the Web are both effective and socially

responsible. Moreover, by bringing together researchers from information retrieval, machine learning, legal informatics, and data governance, the workshop fosters the interdisciplinary dialogue that characterizes The Web Conference. In this sense, ALTARS 2026 extends the conference's mission to explore how the Web can serve as a platform for innovation while safeguarding transparency, inclusiveness, and human oversight.

5 Organizers' Information

All the three organizing committee members have been active participants in the past editions of the TREC, CLEF and FIRE evaluation forum for the Total Recall and Precision Medicine TREC Tasks, TAR in eHealth tasks, and AI for Legal Assistance. We confirm that at least one organizer will attend the conference in person to ensure effective on-site coordination and interaction with participants.

Giorgio Maria Di Nunzio: is Associate Professor at the Department of Information Engineering of the University of Padova. He has been the co-organizer of the ongoing Covid-19 Multilingual Information Access Evaluation forum,⁵ in particular for the evaluation of high-recall systems and high-precision system tasks. He will bring to this workshop the perspective of alternative (to the standard) evaluation measures and multilingual terminological challenges [2, 3, 6, 7, 10].

Evangelos Kanoulas: is Full Professor at the Faculty of Science of the Informatics Institute at the University of Amsterdam. He has been the co-organizer CLEF eHealth Lab and of the Technologically Assisted Reviews in Empirical Medicine task.⁶ He has also been the co-organizer of a related workshop at ECIR 2023, the first international workshop on Legal Information Retrieval (LegalIR). He will bring to the workshop the perspective of the evaluation of the costs in eHealth TAR systems, in particular of the early stopping strategies as well as the conversational aspects of interactive systems [9, 13, 14].

Prasenjit Majumder: is Associate Professor at the Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar and TCG CREST, Kolkata, India. He has been the co-organizer of the Forum for Information Retrieval Evaluation and, in particular, the Artificial Intelligence for Legal Assistance (AILA) task.⁷ He will bring to the workshop the perspective of the evaluation of the costs of eDiscovery, in particular of the issues related to legal precedence findings and text representations in financial texts [1, 8, 12].

6 Program Committee

The Program Committee (PC) for ALTARS 2026 includes researchers and practitioners with expertise in information retrieval, Web data mining, artificial intelligence, and legal and scientific information management.

- Enrico Francesconi, Institute of Legal Informatics and Judicial Systems (IGSG-CNR), Italy
- Udo Kruschwitz, University of Regensburg, Germany
- Wojciech Kusa, TU Wien, Austria
- David Lewis, Nextpoint, USA

- Parth Mehta, Parmonic, USA
- Harrison Scells, University of Tübingen, Germany
- Mark Stevenson, University of Sheffield, UK

References

- [1] Paheli Bhattacharya, Parth Mehta, Kripabandhu Ghosh, Saptarshi Ghosh, Arindam Pal, Arnab Bhattacharya, and Prasenjit Majumder. 2020. FIRE 2020 AILA Track: Artificial Intelligence for Legal Assistance. In *FIRE 2020: Forum for Information Retrieval Evaluation, Hyderabad, India, December 16-20, 2020*, Prasenjit Majumder, Mandar Mitra, Surupendu Gangopadhyay, and Parth Mehta (Eds.). ACM, 1–3. doi:10.1145/3441501.3441510
- [2] Teofan Clipa and Giorgio Maria Di Nunzio. 2020. A Study on Ranking Fusion Approaches for the Retrieval of Medical Publications. *Inf.* 11, 2 (2020), 103. doi:10.3390/info11020103
- [3] Giorgio Maria Di Nunzio and Guglielmo Faggioli. 2021. A Study of a Gain Based Approach for Query Aspects in Recall Oriented Tasks. *Applied Sciences* 11, 19 (2021). doi:10.3390/app11190975
- [4] Giorgio Maria Di Nunzio, Evangelos Kanoulas, and Prasenjit Majumder. 2022. Augmented Intelligence in Technology-Assisted Review Systems (ALTARS 2022): Evaluation Metrics and Protocols for eDiscovery and Systematic Review Systems. In *Advances in Information Retrieval - 44th European Conference on IR Research, ECIR 2022, Stavanger, Norway, April 10-14, 2022, Proceedings, Part II (Lecture Notes in Computer Science, Vol. 13186)*, Springer, 557–560. doi:10.1007/978-3-030-99739-7_69
- [5] Giorgio Maria Di Nunzio, Evangelos Kanoulas, and Prasenjit Majumder. 2023. 2nd Workshop on Augmented Intelligence in Technology-Assisted Review Systems (ALTARS). In *Advances in Information Retrieval - 45th European Conference on Information Retrieval, ECIR 2023, Dublin, Ireland, April 2-6, 2023, Proceedings, Part III (Lecture Notes in Computer Science, Vol. 13982)*, Jaap Kamps, Lorraine Goeuriot, Fabio Crestani, Maria Maistro, Hideo Joho, Brian Davis, Cathal Gurrin, Udo Kruschwitz, and Annalina Caputo (Eds.). Springer, 384–387. doi:10.1007/978-3-031-28241-6_41
- [6] Giorgio Maria Di Nunzio, Stefano Marchesin, and Gianmaria Silvello. 2023. A systematic review of Automatic Term Extraction: What happened in 2022? *Digit. Scholarsh. Humanit.* 38, Supplement_1 (2023), 41–47. doi:10.1093/lc/fqad030
- [7] Giorgio Maria Di Nunzio and Federica Vezzani. 2022. Did I Miss Anything? A Study on Ranking Fusion and Manual Query Rewriting in Consumer Health Search. In *Experimental IR Meets Multilinguality, Multimodality, and Interaction - 13th International Conference of the CLEF Association, CLEF 2022, Bologna, Italy, September 5-8, 2022, Proceedings (Lecture Notes in Computer Science, Vol. 13390)*, Alberto Barrón-Cedeño, Giovanni Da San Martino, Mirko Degli Esposti, Fabrizio Sebastiani, Craig Macdonald, Gabriella Pasi, Allan Hanbury, Martin Potthast, Guglielmo Faggioli, and Nicola Ferro (Eds.). Springer, 217–229. doi:10.1007/978-3-031-13643-6_17
- [8] Surupendu Gangopadhyay and Prasenjit Majumder. 2023. Text representation for direction prediction of share market. *Expert Syst. Appl.* 211 (2023), 118472. doi:10.1016/j.eswa.2022.118472
- [9] Dan Li and Evangelos Kanoulas. 2020. When to Stop Reviewing in Technology-Assisted Reviews: Sampling from an Adaptive Distribution to Estimate Residual Relevant Documents. *ACM Trans. Inf. Syst.* 38, 4 (2020), 41:1–41:36. doi:10.1145/3411755
- [10] Stefano Marchesin, Giorgio Maria Di Nunzio, and Maristella Agosti. 2021. Simple but Effective Knowledge-Based Query Reformulations for Precision Medicine Retrieval. *Inf.* 12, 10 (2021), 402. doi:10.3390/info12100402
- [11] Giorgio Maria Di Nunzio. 2024. Technology Assisted Review Systems: Current and Future Directions. In *Proceedings of the 3rd Workshop on Augmented Intelligence for Technology-Assisted Reviews Systems (ALTARS 2024) co-located with the 46th European Conference on Information Retrieval (ECIR 2024), Glasgow, Scotland, March 28, 2024 (hybrid event) (CEUR Workshop Proceedings, Vol. 3832)*, Giorgio Maria Di Nunzio, Evangelos Kanoulas, and Prasenjit Majumder (Eds.). CEUR-WS.org. <https://ceur-ws.org/Vol-3832/short4.pdf>
- [12] Vedant Parikh, Upal Bhattacharya, Parth Mehta, Ayan Bandyopadhyay, Paheli Bhattacharya, Kripa Ghosh, Saptarshi Ghosh, Arindam Pal, Arnab Bhattacharya, and Prasenjit Majumder. 2021. AILA 2021: Shared task on Artificial Intelligence for Legal Assistance. In *FIRE 2021: Forum for Information Retrieval Evaluation, Virtual Event, India, December 13 - 17, 2021*, Debasis Ganguly, Surupendu Gangopadhyay, Mandar Mitra, and Prasenjit Majumder (Eds.). ACM, 12–15. doi:10.1145/3503162.3506571
- [13] Jie Zou, Mohammad Aliannejadi, Evangelos Kanoulas, Maria Soledad Pera, and Yiqun Liu. 2023. Users Meet Clarifying Questions: Toward a Better Understanding of User Interactions for Search Clarification. *ACM Trans. Inf. Syst.* 41, 1 (2023), 16:1–16:25. doi:10.1145/3524110
- [14] Jie Zou and Evangelos Kanoulas. 2020. Towards Question-based High-recall Information Retrieval: Locating the Last Few Relevant Documents for Technology-assisted Reviews. *ACM Trans. Inf. Syst.* 38, 3 (2020), 27:1–27:35. doi:10.1145/3388640

⁵<http://eval.covid19-mlia.eu>

⁶<https://clefehealth.imag.fr>

⁷<https://sites.google.com/view/aila-2021>